



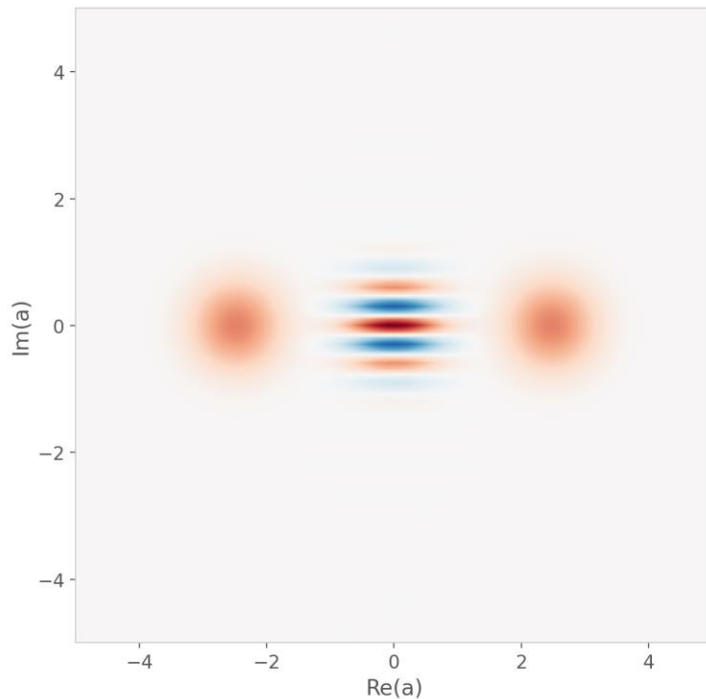
Adaptive Calibration of Cat Qubits

Team: The_μediterranean_operators

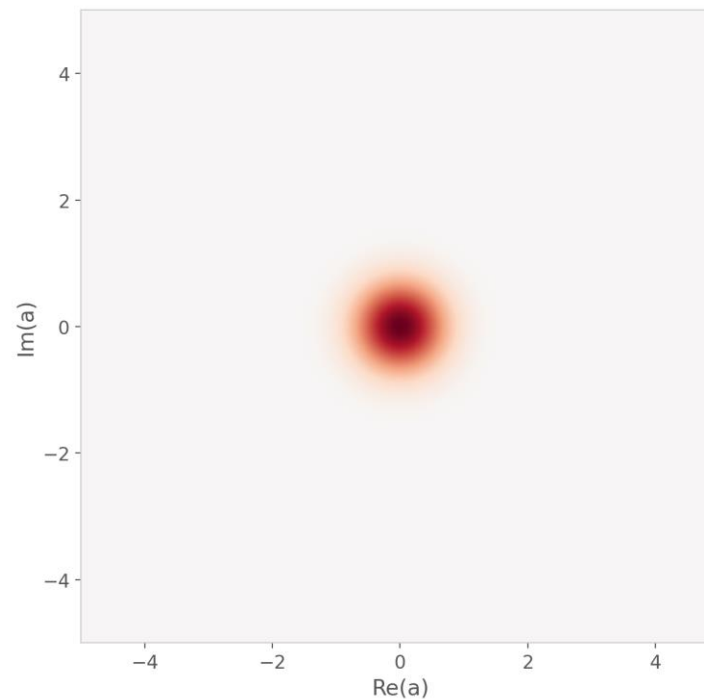
The Challenge: Keeping Cat Qubits Alive



ALIVE — Stabilized Cat (+x state)



DEAD — Vacuum (fully decohered)



Model

- Truncated Hilbertspace:

- Storage mode: $N_A = 12$
- Buffer mode: $N_B = 4$

- Hamiltonian:

$$H = g_2^* a^\dagger a^\dagger b + g_2 a a b^\dagger - \varepsilon_d b^\dagger - \varepsilon_d^* b$$

- Jump operators:

$$L_a = \sqrt{\kappa_a} a$$

$$L_b = \sqrt{\kappa_b} b$$

- Linblad equation:

$$\dot{X} = \frac{i}{\hbar} [H, X] + \sum_i \gamma_i \left(L_i^\dagger X L_i - \frac{1}{2} \{ L_i^\dagger L_i, X \} \right).$$

Model Variants

Model	Knobs	Extra Physics	Use Case
Standard Cat	4	Baseline pumping	Reference
Moon Cat	6	$a^\dagger ab$ coupling	Extended control
Drift-Compensated	4-6	Tunable detuning	Robustness
Kerr Cat	6	Kerr nonlinearity	Higher coherence
Effective Single-Mode	2	Reduced Hilbert space	Fast simulation

Problem Statement

Maximize lifetime of cat qubits under noise and drift.



T_x (Bit-flip lifetime)



$$\eta = T_z / T_x$$



T_z (Phase-flip lifetime)

With balancing control parameters under realistic quantum constraints.

Our Approach



1. Initialize a CMA-ES search distribution around the model's default knob values.



2. Ask for a batch of candidate knob settings from the current distribution.



3. Evaluate each candidate by building the Hamiltonian, simulating the dynamics, and measuring T_z , T_x , and bias.



4. Score each candidate with one of the reward functions defined before.



5. Tell the optimizer the rewards so it updates its sampling distribution.



6. Repeat under the current drift condition, tracking reward, T_z , T_x , and η over epochs.

Reward function objectives

- Log_sum_penalty:

$$R = \ln(T_Z) + \ln(T_X) - \lambda \cdot (\eta - \eta_0)^2$$

- Weighted_log:

$$R = w_Z \cdot \ln(T_Z) + w_X \cdot \ln(T_X) - \lambda \cdot (\eta - \eta_0)^2$$

- Bias_prioritized:

$$R = \ln(T_Z) - \gamma \cdot \max(|\eta - \eta_0| - \varepsilon, 0)$$

- Bias_constrained:

$$R = \ln(T_Z) \text{ if } |\eta - \eta_0| < \varepsilon \text{ else } -\infty$$

- Log_relative_penalty:

$$R = \ln(T_Z) + \ln(T_X) - \lambda \cdot (\ln(\eta) - \ln(\eta_0))^2$$

Computational Optimization

- Running the optimizers with threads on the CPU.
- Build a JAX optimized code for GPU.
- Run our full list of 100 experiments on a cluster of CPUs.

Results



✦ Cat Qubit Optimization Dashboard

The mediterranean operators — ETH Quantum Hackathon 2026 — Alice & Bob Challenge

5 Models

4 Drift Types

5 Penalty Functions

100 Optimization Runs



Cat Qubits

Dissipative cat qubits use two-photon pumping and photon loss to stabilize coherent states $|\alpha\rangle$ and $|\!-\!\alpha\rangle$, offering hardware-efficient quantum error correction.

Two-photon dissipation



The Optimization Problem

Tune control parameters (knobs) to maximize Z and X lifetimes T_Z , T_X while achieving a target bias $\eta = T_Z/T_X$ (e.g. 500:1).

Online CMA-ES



Models

5 cat qubit variants: Standard, Moon, Drift-Compensated, Kerr, and Effective Single-Mode. Each has different knobs and physics.

4–6 knobs each



Hardware Drift

Amplitude and frequency fluctuations (sinusoidal) that degrade performance. Algorithms must adapt online to maintain lifetimes.

Non-stationary

MODEL

Standard Cat

DRIFT

No Drift

PENALTY FUNCTION

Log Sum + Quadratic Bias

► Load

Loaded Standard Cat + No Drift + Log Sum + Quadratic Bias

T_Z (BIT-FLIP)

82.8

T_X (PHASE-FLIP)

0.1650

BIAS H

502

TARGET H

500.0

REWARD

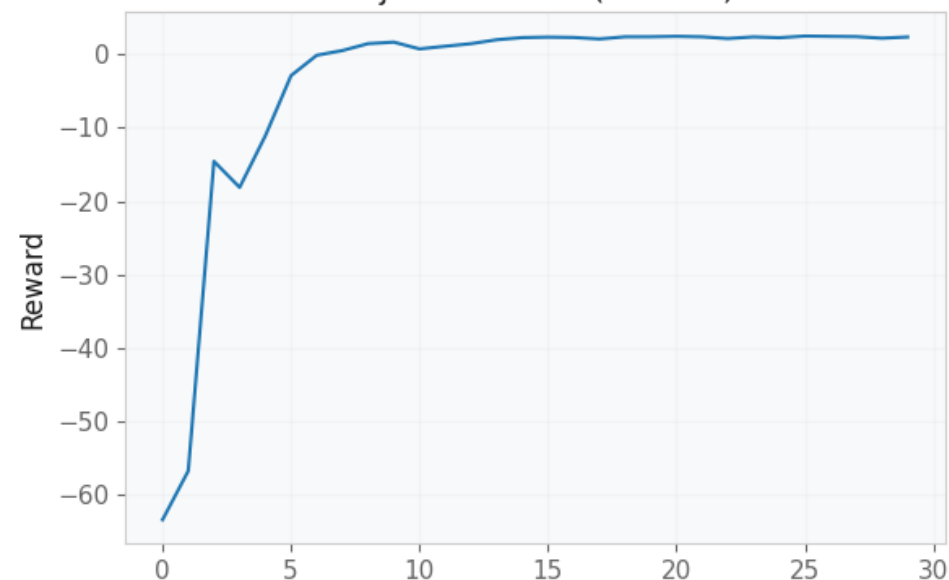
2.41

+ Optimization Graphs

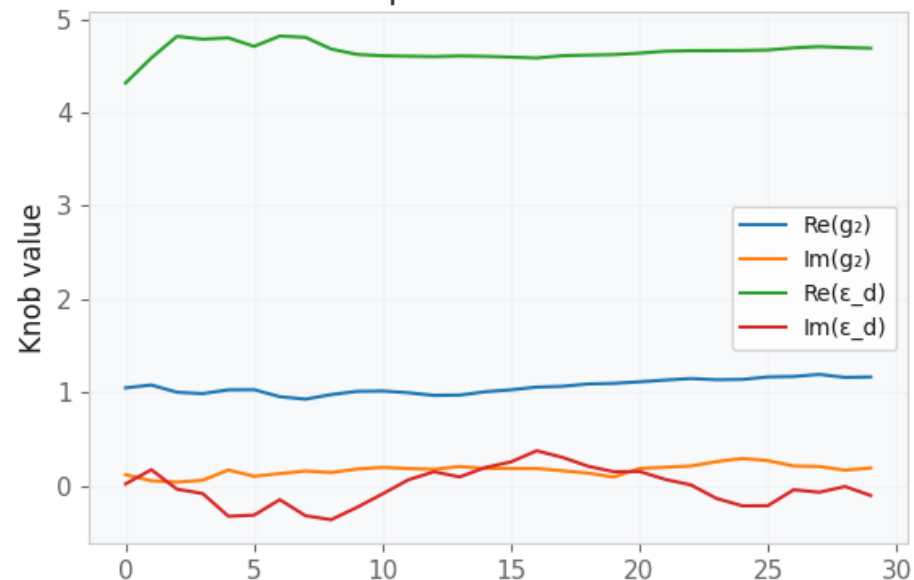
Standard Cat | No Drift | Log sum + quadratic bias penalty

$$R = \ln(T_Z) + \ln(T_X) - \lambda \cdot (\eta - \eta_0)^2$$

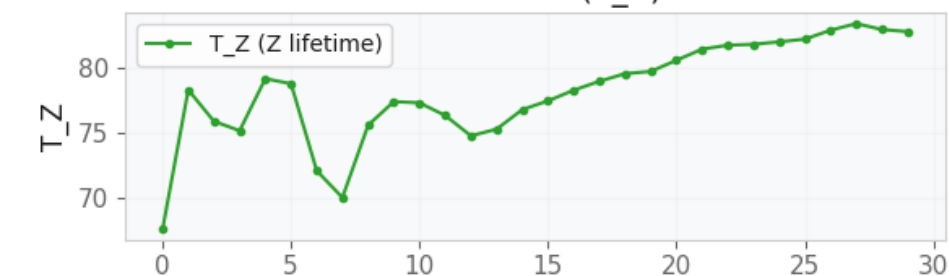
Objective Value (Reward)



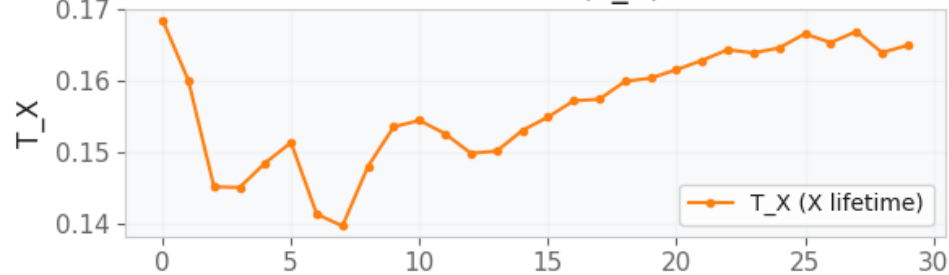
Optimized Knobs



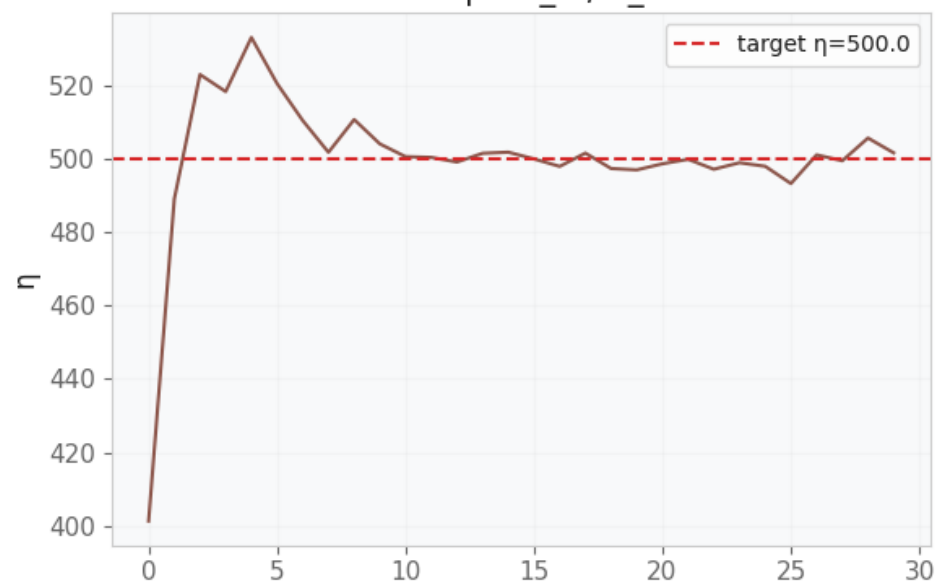
Z Lifetime (T_Z)



X Lifetime (T_X)



Bias $\eta = T_Z / T_X$



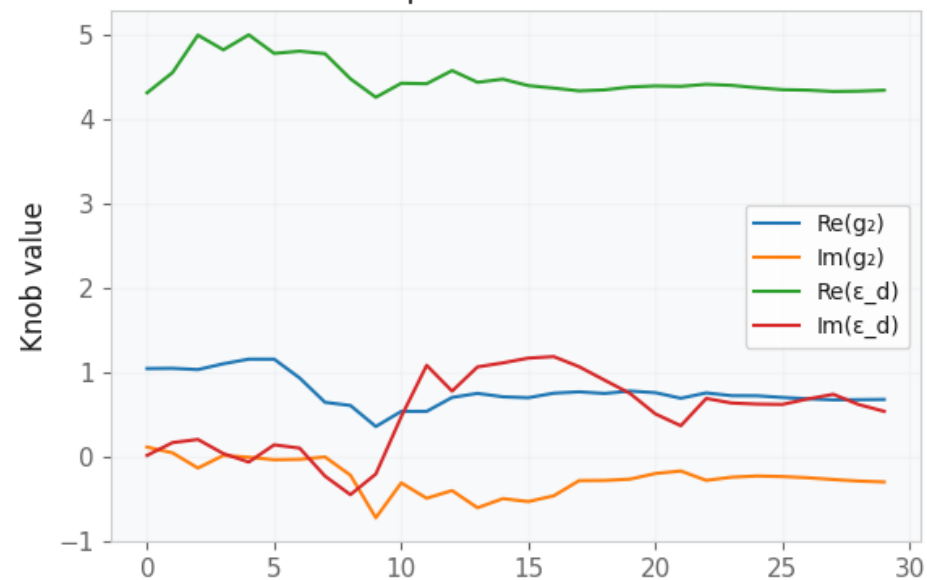
Standard Cat | Frequency Drift | Log sum + quadratic bias penalty

$$R = \ln(T_Z) + \ln(T_X) - \lambda \cdot (\eta - \eta_0)^2$$

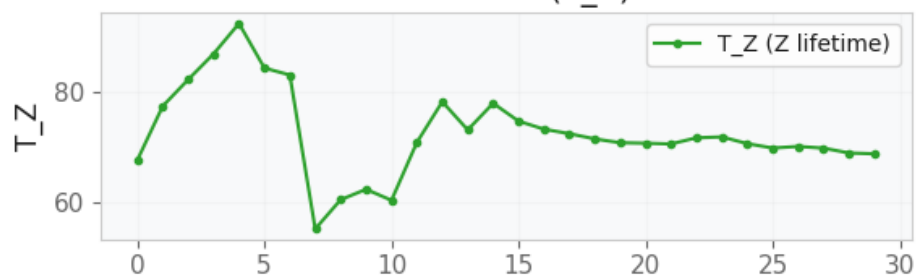
Objective Value (Reward)



Optimized Knobs



Z Lifetime (T_Z)



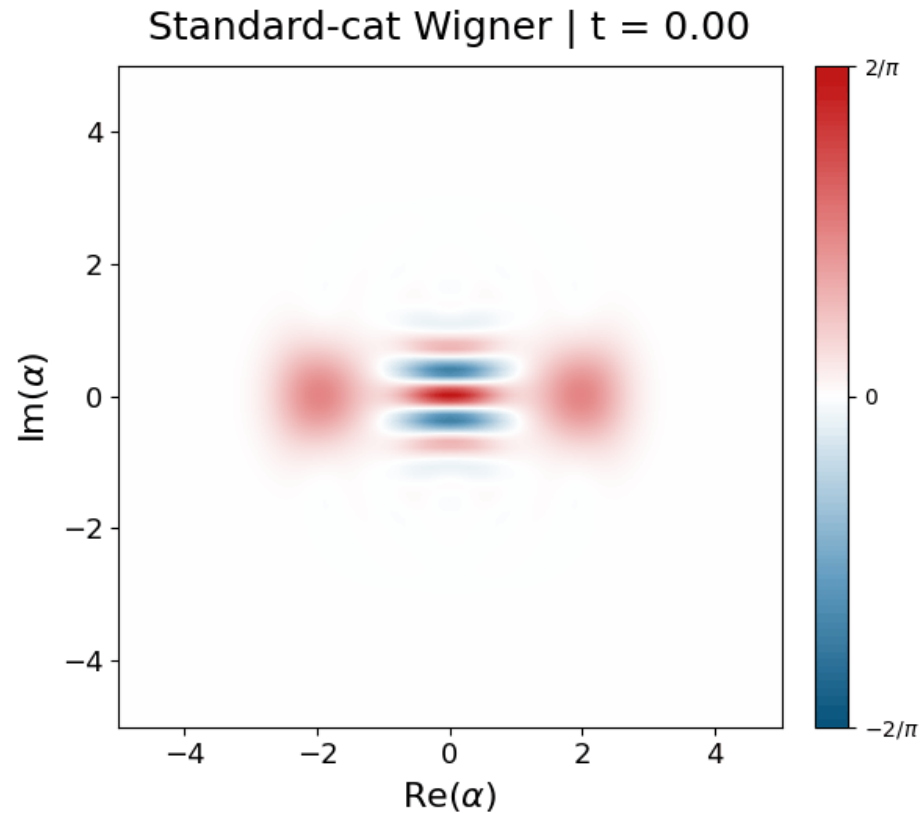
X Lifetime (T_X)



Bias $\eta = T_Z / T_X$



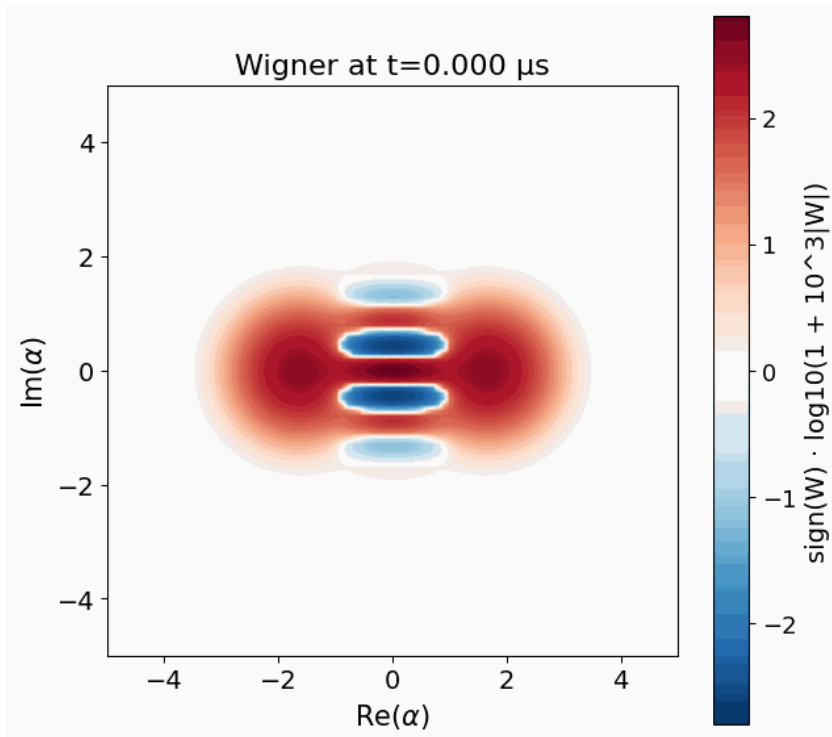
Initialisation



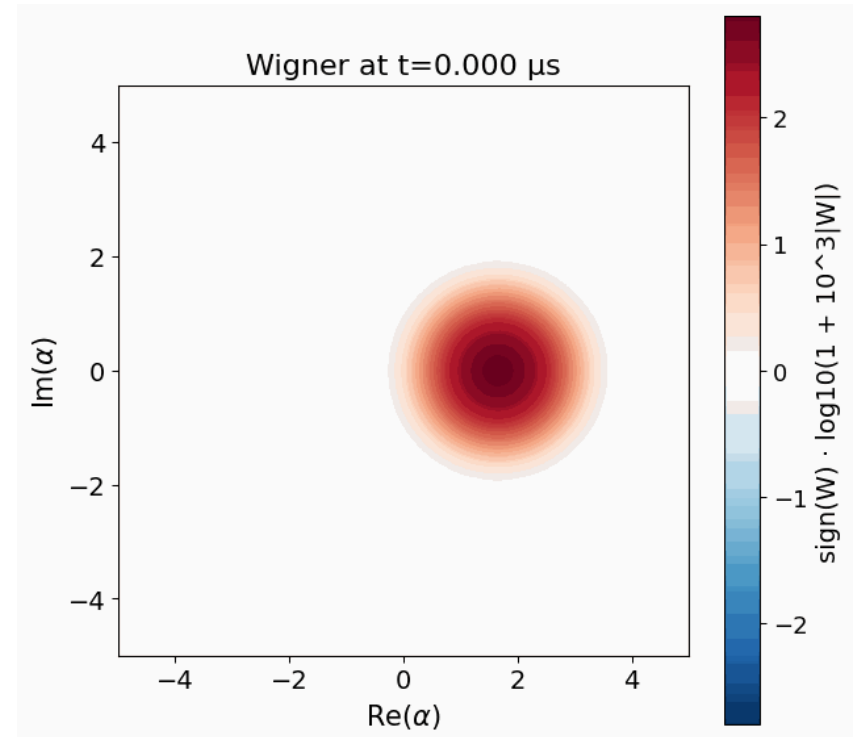
Amplitude Drift

$$H = g_2^* a^{\dagger 2} b + g_2 a^2 b^{\dagger} - \epsilon_d(t) b^{\dagger} - \epsilon_d^*(t) b, \text{ with } \epsilon_d(t) = \epsilon_0 e^{\gamma t}$$

Initial states: +x



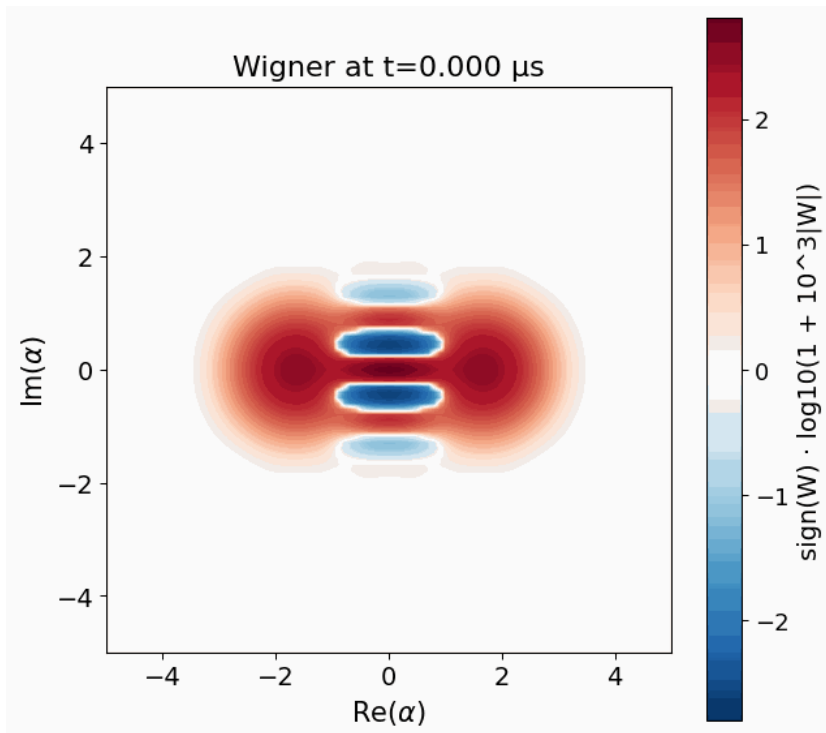
+z



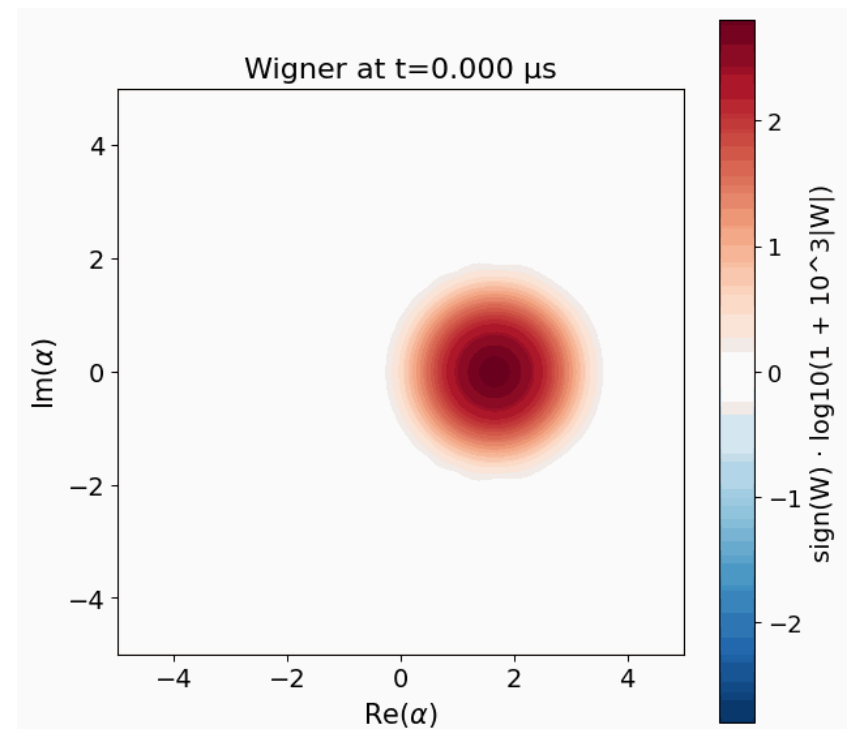
Frequency Drift

$$H = g_2^* a^{\dagger 2} b + g_2 a^2 b^{\dagger} - \epsilon_d b^{\dagger} - \epsilon_d^* b + \Delta(t) a^{\dagger} a, \text{ with } \Delta(t) = A \sin(\omega t + \phi)$$

Initial states: +x



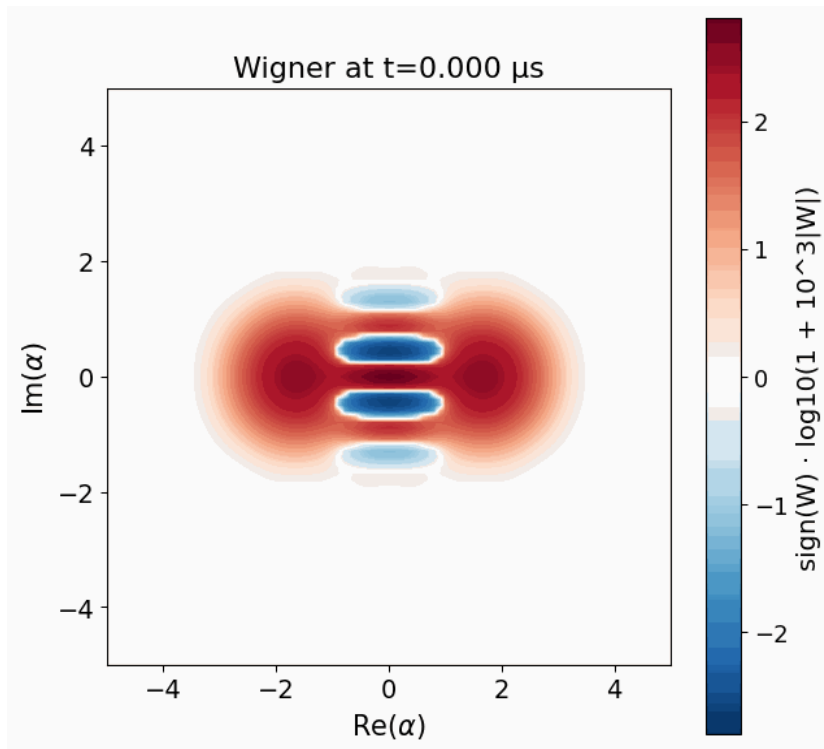
+z



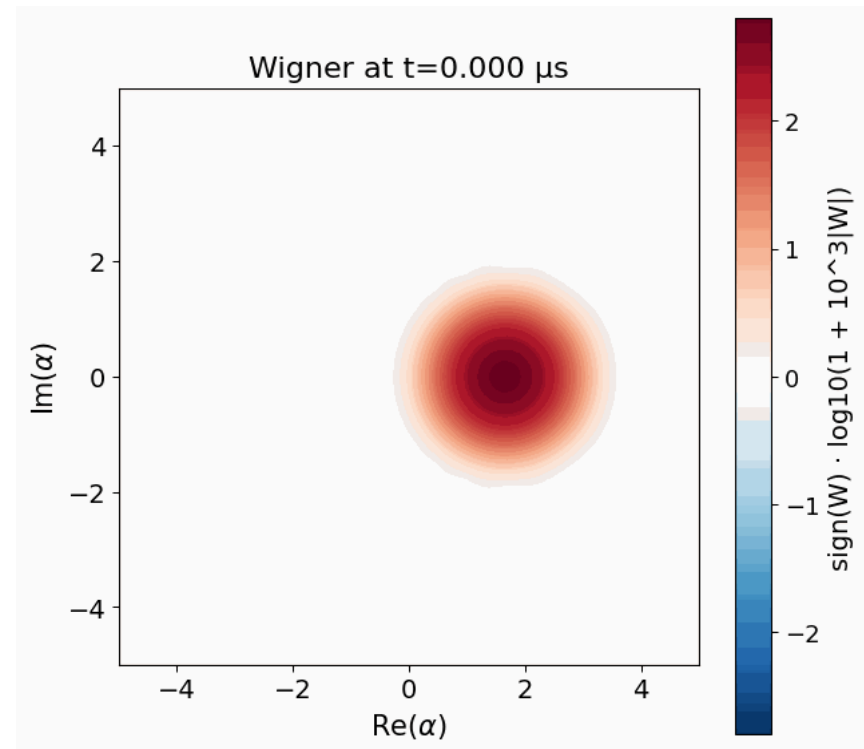
Static Kerr Hamiltonian

$$H = g_2^* a^{\dagger 2} b + g_2 a^2 b^{\dagger} - \epsilon_d b^{\dagger} - \epsilon_d^* b + K (a^{\dagger} a)^2$$

Initial states: +x



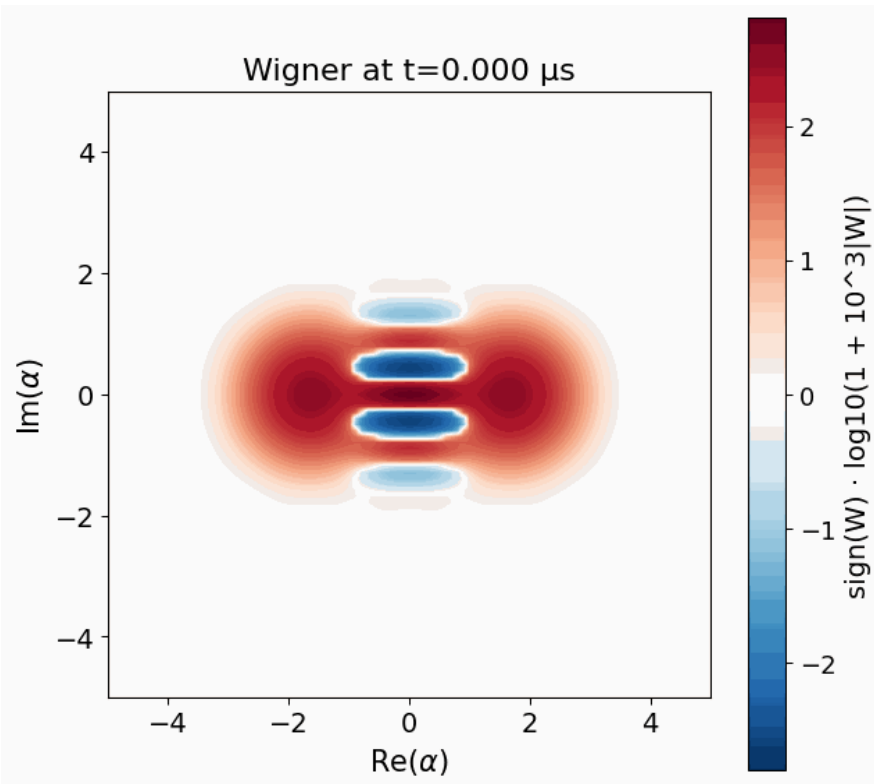
+z



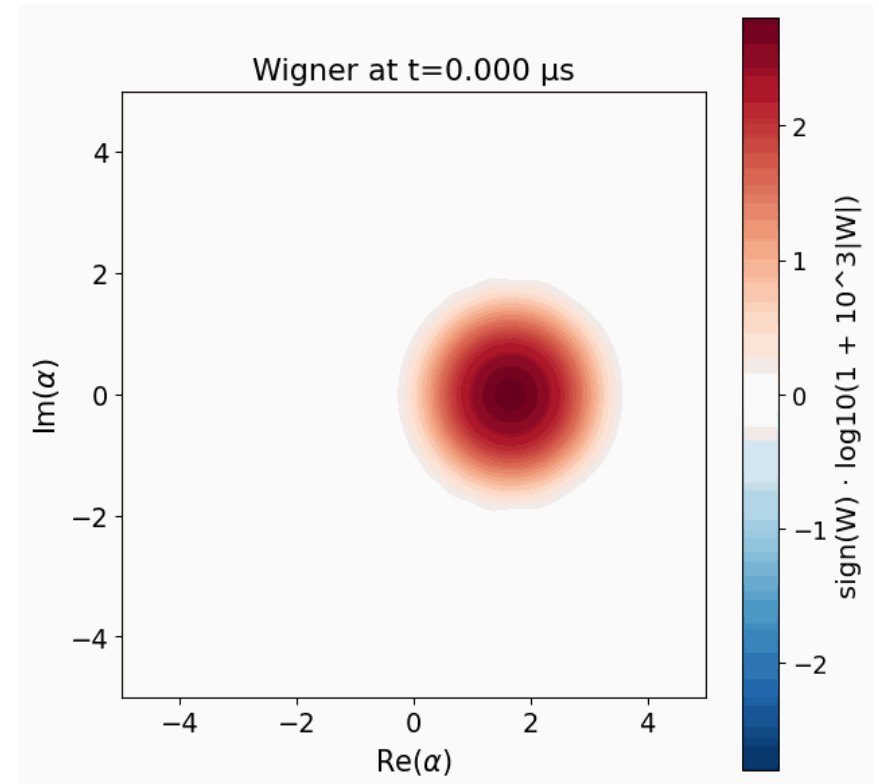
Moon Cat State Hamiltonian

$$H = g_2^* a^{\dagger 2} b + g_2 a^2 b^{\dagger} - \epsilon_d b^{\dagger} - \epsilon_d^* b + g_2 \lambda (a^{\dagger} a - \alpha_{\text{moon}}^2) b + g_2^* \lambda^* (a^{\dagger} a - (\alpha_{\text{moon}}^*)^2) b^{\dagger}$$

Initial states: +x



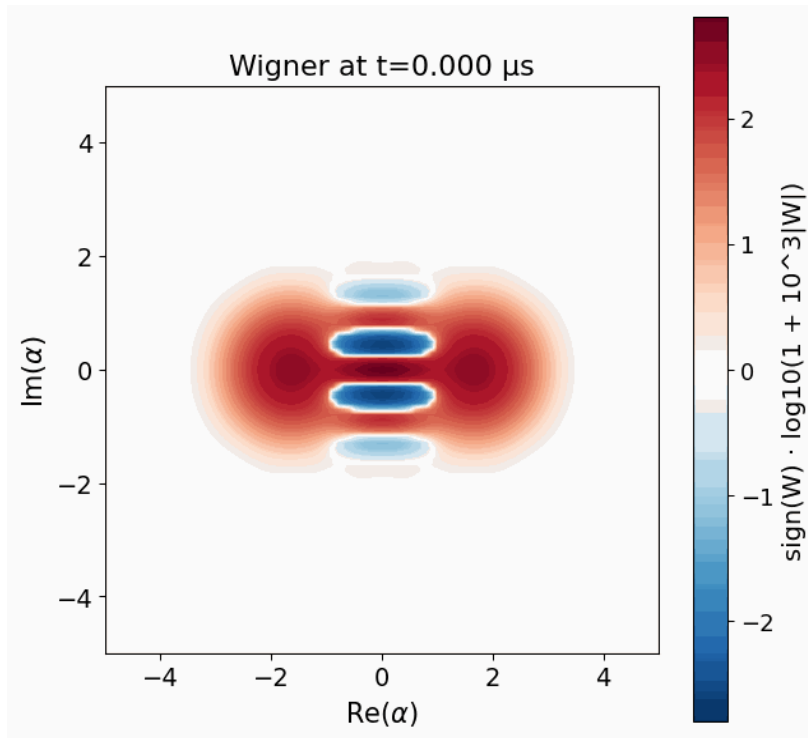
+z



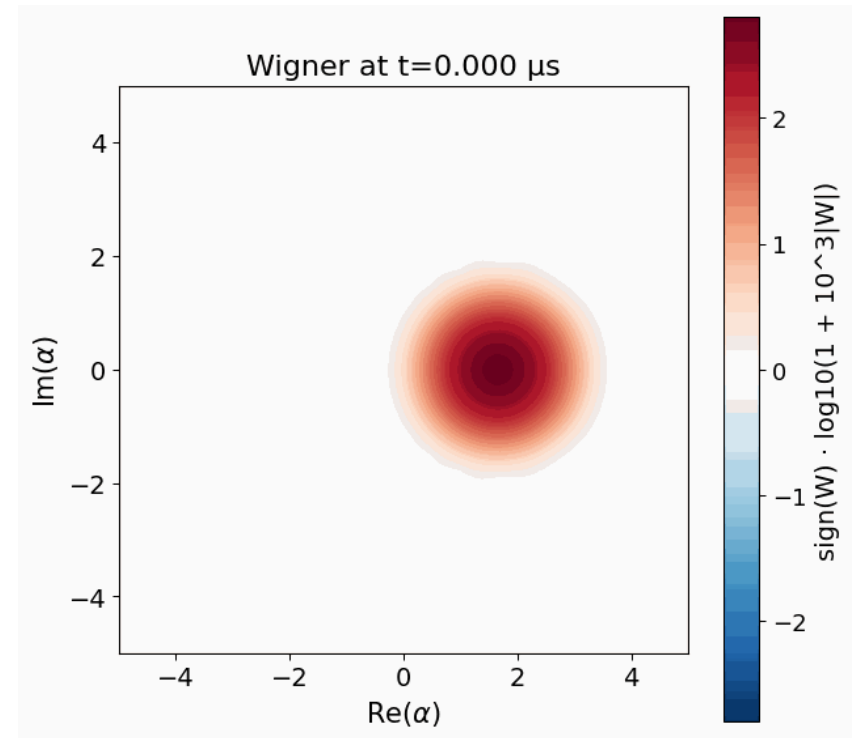
(Original) Cross-Kerr Hamiltonian

$$H = H_{\text{cat}} + \chi a^\dagger a b^\dagger b$$

Initial states: +x



+z



Results



- Improved qubit lifetime
- Stable convergence
- Robust against drift
- Competitive performance vs baselines

Future Work

- Integrate more advanced or different optimizers
- Improve drift modeling and real-time adaptation
- Extend to multi-qubit systems

Conclusion

- Successfully addressed challenge
- Built reusable framework

| Thank you



Any Questions?